

(c) Only 1

(d) Only 2

UPPSC (Pre) 2024

Ans. (c)

'Cereale Yield' is defined as the amount of cereal crops produced per unit area of harvested land. It is generally expressed in kilograms per hectare of harvested land. The cereals measured include wheat, rice, maize, barley, oats, rye millets, sorghum, buckwheat and mixed grains. Therefore, only statement (1) is correct.

101. In the question given below, statement and two assumptions (I) and (II) are given. Choose the correct option from the following :

Statement : In order to replenish the nutrients in the soil, it is important to grow different types of crops in every alternate season.

Assumptions :

(I) A crop can never be grown for the second time in the same field.

(II) If a different crop is grown in the successive season, no additional nutrients such as fertilizers are required to be added to the soil at all.

- (a) Only assumption (I) is implicit.
- (b) Only assumption (II) is implicit.
- (c) Both assumptions (I) and (II) are implicit.
- (d) Neither assumption (I) nor assumption (II) is implicit.

Raj. P.C.S. (Pre) 2024

Ans. (d)

The above statement promotes the periodic change of crops, emphasizing crop diversification (crop rotation) to maintain soil fertility (nutrients). A crop can be grown on the same field again after a few years or within a year interval. Therefore, Assumption (I) is not inherent in the statement. Even when different types of crops are grown in consecutive seasons, fertilizers or manure may still be required to replenish the additional nutrients in the soil. Hence, Assumption (II) is also not inherent in the statement.

102. Which is not an area of Mediterranean type of Agriculture?

- (a) California
- (b) Central Chile

(c) Northern Australia

(d) Southern part of South Africa

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Ans. (c)

The Mediterranean climate is found between 30°-40°/45° latitudes on both sides of the equator, primarily on the western parts of continents. This climatic region is found around the Mediterranean Sea, including Spain, Portugal, France, Southern parts of Italy, Greece, western Turkey, Syria, western Israel and northern Africa. It is also present in California (North America), central Chile (South America), southwestern and southern Australia and the south Western Part of South Africa.

In these regions, Mediterranean type agriculture is practiced, which primarily includes the cultivation of juicy fruits and grapes.

The Minerals

(i) Coal

*When small pieces of rock collect below the surface, it leads to the formation of sedimentary rock also known as layered rocks. Fossils and fossil fuels are found in the sedimentary rocks. Sedimentary rocks occupy 75% of the total Earth's surface. *Minerals such as coal, slate, marble, salt (rock), petroleum are found in sedimentary rocks. Coal and limestone are examples of sedimentary rocks of biotic origin. *Based on the percentage of carbon present coal is categorized into four types—**(1) Anthracite** - It is the best quality of coal having a carbon content of 90%–95%. **(2) Bituminous** – It is next to Anthracite in terms of quality having a carbon content of 70% to 90%. **(3) Lignite** - Famously known as brown coal having a carbon content of 45%–50%. It is mostly used in thermal power plants. **(4) Peat** –It is low-grade coal having a carbon content of 40% or less. **Rock coal** is also known as Anthracite. It is the highest quality of coal. According to the Indian Mineral Yearbook 2022, world proved coal reserves were estimated at 1074108 billion tonnes at the end of 2020 of which 753.639 billion tonnes (70%) have been classified as anthracite & bituminous coal and 320.469 billion tonnes (30%) as sub-bituminous coals & lignite. USA has the largest coal reserves with about 23% share of the total world reserves, followed by Russia (15%), Australia (14%) and China (13%). *According to data from World Mineral Production 2018-2022 in the year 2022, China produced more than half of the world's production of coal. The top five leading producers (2022) of coal in the world are – (1) China, (2) India, (3) Indonesia, (4) USA, (5) Australia. The top 5

countries in terms of proved reserves of coal in the world by the end of year 2020 are USA, Russia, Australia, China and India respectively. *The **North Appalachian** coal area is the largest coal area in the USA. The region produces the best quality of coal. The other important coal regions of USA are – **South Appalachian, Western Interior coal region** and the coal region of Rocky Mountains etc. *Ruhr is a river that flows through Germany. The Ruhr Valley is the main coal region and industrial area known as **Ruhr Basin**. *Saarland is located on the borders of France and Germany in the **Saar River valley**. This area is said to have huge deposits of coal. ***Donets Basin**, also known as Donbass is currently located in Ukraine. It is an important coal producing region. *Provinces of Shanxi, Shensi and Sichuan in China are important coal producing areas. Kuznets Basin of Russia and Lancashire in the U.K. are also famous for coal production.

1. Coal is an example of –

- (a) Igneous (b) Metamorphic rock
(c) Sedimentary (d) All of the above

Uttarakhand P.C.S. (Pre) 2006

Ans. (c)

Sedimentary rocks are formed by the deposition of sediment usually in the form of layers. Sedimentary rocks are also known as layered rocks. Most fossils and mineral oil are found in these rocks. Sedimentary rocks cover 75% of the Earth's surface. Minerals like Coal, Slate, Limestone, Salt, Petroleum, Marble, etc. are examples of sedimentary Rock.

Depending on the amount of carbon and hardness, coal is Segregated in to four categories –

- (i) Anthracite : It is the best kind of coal. It has about 90-95% carbon content.
(ii) Bituminous : It is the second most important type of coal and is the most abundant coal. It has a carbon content between 70-90%.
(iii) Lignite: It is often referred to brown coal. It has a carbon content between 45-50%. It is mostly used in thermal power stations.
(iv) Peat : It is the lowest rank of coal having a carbon content of less than 40%.

2. Coal, Crude Oil and Natural gas are called –

- (a) Crude Fuel (b) Traditional Fuel
(c) Natural Fuel (d) Fossil Fuel

Uttarakhand P.C.S. (Mains) 2002

Ans. (d)

Coal, Crude oil and Natural gas are Fossil Fuels. Fossil fuels were formed in the geological past over the course of millions of years from the remains of living organisms.

3. Which of the following statements are correct :

- I. China is the leading coal producer of the World**
II. Donetsk Basin is the major coal producing region in Ukraine
III. Saar Region is the major coal producing area in Germany
IV. The major coal mining areas of the USA lie in the Appalachian Region

Select the correct answer from the code given below :

Code :

- (a) I, and II (b) II and III
(c) I, II and III (d) All

U.P.P.C.S. (Pre) 1999

Ans. (d)

China ranks first in the production of coal. Thus Statement-I is correct.

Donetsk Basin, also known as Donbass, is situated in Ukraine and is a major coal producing region. Thus, statement II is correct.

Saar Region is a forest in the German state bordered by France. A large deposit of coal is found here. Thus, statement III is correct.

The biggest coal region in the United States of America is North Appalachian coal region. The best quality coal is found here. In addition to this, the other coal regions in the USA are Western Interior coal region, coal region of Rocky Mountains, etc. Thus statement IV is also correct.

4. 'Donbass' region is famous for –

- (a) Iron ore (b) Coal
(c) Copper ore (d) Gold

U.P.P.C.S. (Pre) 2007

Ans. (b)

See the explanation of the above question.

5. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I (Coal fields)	List-II (Countries)
A. Donetsk	1. Germany
B. Kuznetsk	2. U.K
C. Lancashire	3. Russia
D. Saar	4. Ukraine

Code :

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 1 | 2 | 3 | 4 |
| (b) | 4 | 3 | 2 | 1 |
| (c) | 3 | 4 | 1 | 2 |
| (d) | 1 | 3 | 2 | 4 |

U.P.P.C.S. (Pre) 2013

Ans. (b)

The correctly matched list of coal fields with their countries is as follows:

(Coalfields)	(Countries)
Donetsk	Ukraine
Kuznetsk	Russia
Lancashire	U.K.
Saar	Germany

6. Which of the following countries is the leading country in the reserve of Hard Coal?

- (a) Nepal (b) China
(c) New Zealand (d) India

40th B.P.S.C. (Pre) 1995

Ans. (b)

China was at the top in terms of the reserve of hard coal when this question was asked followed by the USA and India. According to the Indian Minerals Yearbook 2022, data the USA ranks first in the proved reserve of coal at the end of 2020 followed by Russia, Australia, China & India respectively.

7. Which one of the following is the largest coal producing centre of Ukraine?

- (a) Zaporizhzhia (b) Kharkiv
(c) Kyiv (d) Donetsk

U.P.P.C.S. (Pre) 2023

Ans. (d)

Coal mining in Ukraine is often associated with the coal-rich Donetsk basin. However this is not the only coal mining region, others include the Lviv-Volhynian basin and the Dnieper brown coal mining basin.

The largest coal-producing centre in Ukraine is the Donetsk Basin, also known as the Donbas region. This region is located in the eastern part of Ukraine, near the border with Russia.

The Donbas region produces about 90% of Ukraine's total coal output and is home to numerous large coal mines.

8. More than half of the world's production of coal comes from :

- (a) U.S.A. and Russia (b) China and U.S.A.
(c) China and India (d) South Africa and Australia

U.P.P.C.S. (Mains) 2006

Ans. (*)

Option (b) is the correct answer when the question is asked. According to data from World Mineral Production 2018-22 in the year 2022, China produced more than half of the world's of coal.

9. The Ruhr basin is the famous industrial region of –

- (a) China (b) Germany
(c) Japan (d) United Kingdom

U.P.P.C.S (Pre) 2010

Ans. (b)

Ruhr is a river in Germany. Ruhr Basin is in Western Germany. Some of the important coal mines and industrial areas are situated in the valley, known as the 'Ruhr Basin'.

10. Famous 'Ruhr coalfield' is situated in which of the following countries?

- (a) Germany (b) France
(c) Great Britain (d) Russia

U.P.P.C.S. (Pre) 2022

Ans. (a)

Ruhr coalfield is situated in Germany. The Ruhr coalfield (extending west of the Rhine and north of the Lippe) is one of the world's largest, producing the bulk of Germany's bituminous coal. Steel production and diversified chemical manufacturing constitute the other basic industries of the region, which is served by an extensive inland-waterway system and one of Europe's densest railway networks. Hence option (a) is correct.

11. Which of the following countries is the largest reservoir of coal?

- (a) Australia (b) China
(c) New Zealand (d) Brazil

M.P.P.C.S. (Pre) 2010

Ans. (b)

Among the given options, China is the largest reservoir and producer of coal when the question is asked. According to the Indian Minerals Yearbook 2022, Australia has the leading position in the largest coal reserves among the options given above. The top three coal reserves globally are the USA, Russia and Australia.

12. The highest coal-producing country in the world is-

- (a) India (b) USA
(c) China (d) Russia

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

According to World Mineral Production 2018-22, China ranks first in coal production in the year 2022.

13. Rukwa lake region (Tanzania) is famous for which of the following mineral?

- (a) Mica (b) Coal
(c) Iron ore (d) Gold

U.P.U.D.A./L.D.A. (Spl) (Mains) 2010

Ans. (b)

Various stones suitable for ornaments are found in the Rukwa lake region (Tanzania). Emerald, Moonstone, Aquamarine, Amethyst, Ruby and Topaz are prominent among them. Some coal layers are found in this region and some coal mining work has also been done here. Hence, option (b) is the correct answer.